Exercise 2.22. Suppose that a, b and c are positive integers, and $\gcd(a,b) = d$. Prove that $a \mid b$ if and only if $d = a$ To do this, here are the two things that you should prove:

(a) If $a \mid b$, then $d = a$.

(b) If $d = a$, then $a \mid b$.

Solution.

Let a, b, c be positive integers, and suppose

$$\gcd(a,b) = d.$$

We want to prove

$$a \mid b \iff d = a.$$

(a) If $a \mid b$, then $d = a$

If $a \mid b$, then a divides both a and b. Thus a is a common divisor of a and b.

Since $d = \gcd(a,b)$ is the **greatest** common divisor, we have

$$a \mid d.$$

Also, because $d = \gcd(a,b)$, $d \mid a$. Therefore,

$$a \mid d \quad \text{and} \quad d \mid a.$$

Since $a, d > 0$, it follows that

$$\boxed{d = a}.$$

(b) If $d = a$, then $a \mid b$

By definition of the greatest common divisor,

$$d \mid b.$$

If $d = a$, substitute a for d:

$$\boxed{a \mid b}.$$

Therefore,

$$\boxed{a \mid b \text{ if and only if } \gcd(a,b) = a}.$$